Module 3: Exchange and transport

In this module, learners study the structure and function of gas exchange and transport systems in a range of animals and in terrestrial plants.

The significance of surface area to volume ratio in determining the need for ventilation, gas exchange and transport systems in multicellular organisms is emphasised. The examples of terrestrial green plants and a range of animal phyla are used to illustrate the principle.

Learners are expected to apply knowledge, understanding and other skills developed in this module to new situations and/or to solve related problems.

3.1 Exchange and transport

3.1.1 Exchange surfaces

As animals become larger and more active, ventilation and gas exchange systems become essential to supply oxygen to, and remove carbon dioxide from, their bodies.

Ventilation and gas exchange systems in mammals, bony fish and insects are used as examples of the properties and functions of exchange surfaces in animals.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the need for specialised exchange surfaces	To include surface area to volume ratio (SA:V), metabolic activity, single-celled and multicellular organisms. $\text{ratio} = \frac{\text{surface area}}{\text{volume}}$ *M0.1, M0.3, M0.4, M1.1, M2.1, M4.1* HSW1, HSW3, HSW5, HSW8
(b)	the features of an efficient exchange surface	To include: • increased surface area – root hair cells • thin layer – alveoli • good blood supply/ventilation to maintain gradient – gills/alveolus.
(c)	the structures and functions of the components of the mammalian gaseous exchange system	To include the distribution and functions of cartilage, ciliated epithelium, goblet cells, smooth muscle and elastic fibres in the trachea, bronchi, bronchioles and alveoli. **PAG1** HSW8
(d)	the mechanism of ventilation in mammals	To include the function of the rib cage, intercostal muscles (internal and external) and diaphragm. HSW8

(e)	the relationship between vital capacity, tidal volume, breathing rate and oxygen uptake	To include analysis and interpretation of primary and secondary data e.g. from a data logger or spirometer. *M0.1, M0.2, M0.4, M1.3* **PAG10** HSW2, HSW3, HSW4, HSW5, HSW6
(f)	the mechanisms of ventilation and gas exchange in bony fish and insects	To include: • bony fish – changes in volume of the buccal cavity and the functions of the operculum, gill filaments and gill lamellae (gill plates); countercurrent flow • insects – spiracles, trachea, thoracic and abdominal movement to change body volume, exchange with tracheal fluid. HSW8
(g)	the dissection, examination and drawing of the gaseous exchange system of a bony fish and/or insect trachea	**PAG2** HSW4
(h)	the examination of microscope slides to show the histology of exchange surfaces.	**PAG1** HSW4